Average rates of change

1

a -4 **b** -3 **c** -7 **d** Variable

2 Constant

3

a -1 **b** -1 **c** -1 **d** Linear

4

a -4 **b** -4 **c** -4 **d** Linear

5

a Constant **b** Variable **c** Constant **d** Variable

e Variable **f** Variable **g** Variable **h** Variable

6

a $\frac{23}{4}$ **b** 5 **c** $\frac{23}{5}$ **d** Non-linear

7 Non-linear

8

a 1

b 2

c 4

d

Interval	$[3,4]$	$[4,5]$	$[5,6]$
Average rate of change	8	16	32

9

a $\frac{9}{10}$

b The cost of a call increases by $\$0.90$ for each additional minute.

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10

a 500

b Aside from the pre-ordered tickets, there were 500 tickets purchased each day.

11

a $-\frac{1}{250}$

b The price of each toy decreases by $\$1.00$ for each additional 250 units in supply.

12

a 0

b The number of earthquakes per year is a constant value.

Applications

13

a $y = 11$ **b** $y = 26$ **c** $\frac{15}{4}$

14 272

15 $\$300$ per year

16 60

17 6.5 insects per week

18 1.6

19

a

i $9\ \text{ft/s}$ **ii** $3\ \text{ft/s}$ **iii** $-3\ \text{ft/s}$ **iv** $-9\ \text{ft/s}$

b Between $t = 0$ and $t = 1$, and between $t = 3$ and $t = 4$

c The ball is falling down.

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20

a

Year	0	1	2	3	4	5
Tobia's staff	310	269	228	187	146	105
Gwen's staff	310	302	286	262	230	190

b -41

c -32

d Linear, because it decreases by the same amount each year.

e Non-linear, because it decreases by more each year.

f Tobias's

g Gwen's

21

a

Month	0	1	2	3	4	5
Mario's clients	0	27	54	81	108	135
Christa's clients	0	6	18	36	60	90

b Linear

c Non-linear

d 27

e 18

f Mario's

g Christa's

22

a

Month	0	1	2	3	4	5
Value of Harry's investment	2000	2060	2120	2180	2240	2300
Value of Skye's investment	2000	2040	2080.8	2122.42	2164.87	2208.17

b \$60

c \$40.8

d Linear

e Non-linear

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23

a $y = 4x + 20$

b $y = 20\,(1.10)^x$

c 4 turbines each year

d 9.59 turbines each year

e Option A

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